

# Coupling Effects of CO<sub>2</sub> and Thermal Treatment on the Rheological Behavior of Heavy Oil: An Integrated Experimental and Modeling Study

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Cite This: <https://doi.org/10.1021/acsomega.5c09369>



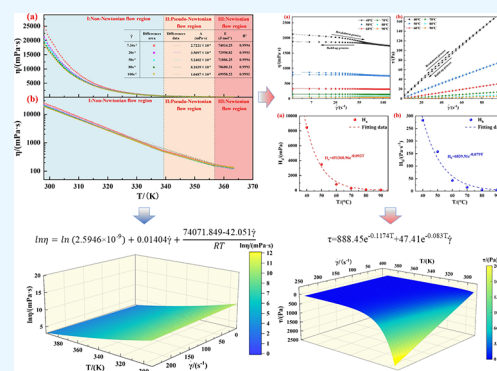
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**ABSTRACT:** The rheological behavior of high-viscosity heavy oil is crucial for its efficient development. CO<sub>2</sub>-assisted thermal recovery serves as an effective method to enhance heavy oil mobility. However, existing studies still lack sufficient quantitative characterization of the coupling effect of thermal and CO<sub>2</sub> interactions on improving heavy oil flow capacity. To address this issue, this study thoroughly investigates the synergistic viscosity reduction mechanism and the evolution of rheological properties during heavy oil extraction under combined CO<sub>2</sub> and thermal effects. Through systematic rheological testing and theoretical modeling, a modified Arrhenius model incorporating a shear correction factor was developed, enabling accurate prediction of the viscosity–temperature relationship under different shear rates. Dual hysteresis loop analysis was employed to quantify the effects of thermal and shear history, confirming that the thixotropic recovery capability of heavy oil is governed by both thermal and shear history and revealing an exponential decay pattern of thixotropic strength with increasing temperature and shear rate. A temperature-dependent Bingham constitutive equation was established, achieving precise prediction of rheological behavior across the full temperature range from the non-Newtonian to the Newtonian regime. This study elucidates the spatiotemporal evolution of heavy oil rheological behavior throughout the entire CO<sub>2</sub>-thermal synergistic extraction process, providing a key theoretical tool for accurate prediction of development performance and dynamic regulation of production parameters.



## 1. INTRODUCTION

The global geological reserves of heavy oil and oil sands exceed 8 trillion barrels,<sup>1–4</sup> far surpassing those of conventional crude oil, making them a crucial strategic alternative resource for future energy supply. As a typical non-Newtonian fluid, improving the rheological properties of heavy oil is the core objective of enhancing its recovery.

At present, thermal recovery remains the primary method for heavy oil development. By injecting thermal fluids such as steam into the reservoir to increase formation temperature, the viscosity of heavy oil is reduced, thereby improving its flowability.<sup>5–7</sup> As early as 1966, Trintopec conducted the first thermal recovery experiment on heavy oil,<sup>8</sup> carrying out a small-scale cyclic project test in the Palo Seco field, which demonstrated the significant role of thermal effects in enhancing heavy oil flowability. Kumar P.<sup>9</sup> conducted rheological experiments in the temperature range of 35–65 °C using oscillatory tests (OSC), universal control rate (CR), and controlled stress (CS) modes, confirming that heavy oil exhibits non-Newtonian fluid characteristics at low temperatures, which diminish as temperature increases. Ilyin S.O.<sup>10</sup> investigated the effects of temperature and shear on the

rheological properties of heavy oil, showing that the complex molecular structure of heavy oil is responsible for its high viscosity and yield stress. High temperature and high shear can reduce the viscosity of heavy oil, while the addition of light hydrocarbon solvents can suppress its yield stress. Mario R.<sup>11</sup> proposed a viscosity calculation model based on the Arrhenius equation for heavy oil and light oil binary mixtures, applicable for viscosity prediction across a wide temperature range. Although extensive research has been conducted on the impact of thermal effects on the rheological properties of heavy oil, most heavy oil reservoirs worldwide have now entered the late stages of steam injection,<sup>12–14</sup> facing challenges such as low thermal efficiency, significant heat loss, and high development costs. These limitations are particularly pronounced in ultradeep heavy oil reservoirs.<sup>15,16</sup> Therefore, there is a need

**Received:** September 9, 2025

**Revised:** November 14, 2025

**Accepted:** November 17, 2025

Table 1. SARA Components and Basic Properties of Heavy Crude Oils

| Oil type  | Density (g·mL <sup>-1</sup> , 20 °C) | Viscosity (mPa·s, 50 °C) | SARA components(wt.%) |          |       |            |
|-----------|--------------------------------------|--------------------------|-----------------------|----------|-------|------------|
|           |                                      |                          | Saturate              | Aromatic | Resin | Asphaltene |
| Heavy oil | 0.927                                | 2847.64                  | 51.7                  | 18.11    | 16.98 | 5.28       |

to introduce auxiliary methods to enhance heavy oil flowability and improve recovery. In recent years, CO<sub>2</sub>-assisted thermal recovery technology has garnered widespread attention due to its unique technical advantages.<sup>17,18</sup> CO<sub>2</sub> exhibits good solubility in heavy oil and can significantly improve its flowability through multiple mechanisms.<sup>19–22</sup> Compared to traditional steam injection processes, CO<sub>2</sub>-assisted steam processes require much less thermal energy, reducing the temperature of the injected fluid.<sup>23,24</sup> Srivastava R.K.<sup>25</sup> evaluated the applicability and effectiveness of CO<sub>2</sub> for heavy oil recovery, showing that dissolved CO<sub>2</sub> causes crude oil to swell, reduces oil phase density, and significantly decreases crude oil viscosity. Simultaneously, CO<sub>2</sub> can extract light components from crude oil, improving oil phase composition and effectively enhancing heavy oil flowability. Geng H.Z.<sup>26</sup> tested the influence of CO<sub>2</sub> on heavy oil viscosity, confirming that the viscosity of the oil-gas mixture decreases with increasing pressure until the bubble point pressure is reached. Lu N.<sup>27</sup> systematically explored the mechanisms of enhanced oil recovery in heavy oil through microscopic visualization experiments and molecular dynamics simulations in a mixed CO<sub>2</sub> thermal system. The results indicated strong van der Waals forces between CO<sub>2</sub> and heavy oil molecules, and the addition of CO<sub>2</sub> effectively promotes the decomposition and swelling of heavy oil. CO<sub>2</sub> can significantly reduce interactions between heavy oil and rock surfaces as well as within the heavy oil itself. Hu R.E.<sup>28</sup> systematically studied the influence patterns on heavy oil rheological capacity, proposing that the viscosity-temperature relationship of the mixed fluid still follows an exponential decline pattern, and that CO<sub>2</sub>-dissolved heavy oil still exhibits non-Newtonian fluid characteristics. Zhang Y.<sup>29</sup> explored the nucleation and growth patterns of CO<sub>2</sub> bubbles in CO<sub>2</sub>-heavy oil systems, demonstrating that the structure of asphaltenes in heavy oil promotes the formation of CO<sub>2</sub> cavities, thereby enhancing the flowability of the mixed fluid. Most importantly, the synergistic effect between CO<sub>2</sub> and thermal energy can significantly enhance viscosity reduction. Field tests have shown that CO<sub>2</sub>-assisted steam flooding can substantially improve recovery rates, demonstrating promising application prospects.

However, current understanding of the rheological behavior of heavy oil under the synergistic action of CO<sub>2</sub> and heat remains significantly inadequate. Particularly under the multi-field coupling conditions of temperature, shear, and CO<sub>2</sub> dissolution, the evolution patterns of the microscopic structure and the macroscopic rheological response characteristics of heavy oil are still unclear. Traditional Arrhenius models only consider the influence of temperature on viscosity and fail to account for the effect of shear rate, leading to deviations between predicted results and actual data. Furthermore, the impact of thermal history experienced by heavy oil during thermal recovery on its rheological properties lacks systematic study. These knowledge gaps severely constrain the precise design and optimization of CO<sub>2</sub>-assisted thermal recovery technologies.

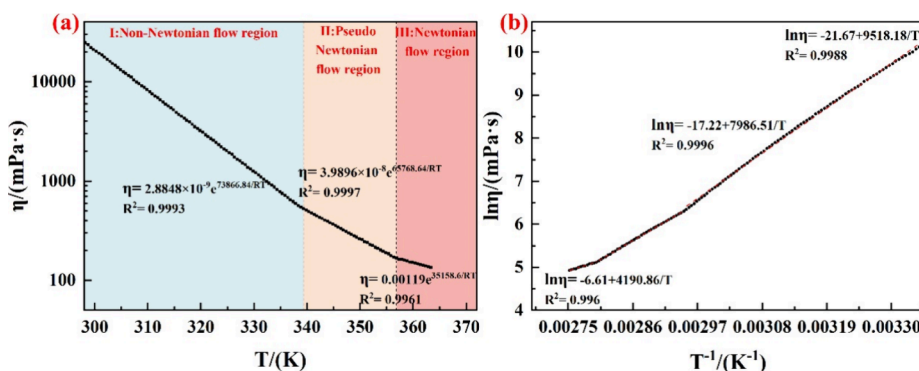
In response to the aforementioned challenges, this study employs a combined approach of systematic experimentation

and theoretical modeling to thoroughly investigate the influence mechanism of CO<sub>2</sub> dissolution on the rheological behavior of heavy oil. The research focuses on the coupling effects of temperature, shear, and CO<sub>2</sub> dissolution, utilizing rheometric testing methods to reveal the evolution of rheological properties of heavy oil under the synergistic action of CO<sub>2</sub> and heat. The phase transition critical conditions for the rheological behavior of heavy oil were determined, a viscosity-temperature prediction model considering shear effects was established, the viscosity reduction mechanism of the CO<sub>2</sub>-thermal synergistic effect was clarified, and a temperature-dependent constitutive equation was constructed to accurately characterize the rheological behavior of heavy oil across the entire temperature range. This study provides deeper insights into the rheological behavior of heavy oil under CO<sub>2</sub>-thermal synergy and establishes a viscosity prediction method under multifield coupling conditions. The findings are expected to advance the innovation of CO<sub>2</sub>-assisted thermal recovery technology, with significant practical implications for enhancing heavy oil development efficiency and economic returns.

## 2. EXPERIMENTAL SECTION

**2.1. Experimental Samples.** The experimental oil samples were obtained from Block J230 in the Xinjiang Oilfield. The crude oil underwent dehydration and degassing treatment, and its basic physical properties are shown in Table 1. The main components of this crude oil are saturates (51.7%), while the content of resins and asphaltenes exceeds 20%, classifying it as a typical high-viscosity heavy oil. To eliminate the "memory effect" of historical thermal and shear actions on the colloidal structure, all samples were maintained at a constant temperature of 90 °C for 60 min and then quenched to the initial experimental temperature. Additionally, to ensure experimental accuracy, the crude oil was subjected to pretreatment operations such as filtration and impurity removal to eliminate interference from other factors. The gas used in the experiments was CO<sub>2</sub> with a purity ≥ 99.99%.

**2.2. Viscosity-Temperature Characteristics of Heavy Oil.** The viscosity measurements were conducted following the Chinese National Standard GB/T 28910–2012, which employs a rotational viscometer to determine the apparent viscosity of crude oil through controlled shear rate testing under precisely regulated temperature conditions. The rheological measurements were performed using an MCR 302 rheometer (Anton Paar, Austria), with an operable temperature range of 90–300 °C and a maximum pressure tolerance of 15 MPa. The temperature test range was set from 25 to 95 °C with a heating rate of 0.5 °C/min. Shear rates were set at 7.34, 20, 50, 75, and 100 s<sup>-1</sup> respectively to investigate the viscosity variation under different shear conditions. The testing procedure consisted of heating the sample from 25 to 90 °C at a fixed heating rate of 0.5 °C/min, followed by immediate cooling from 90 °C back to 25 °C at the same rate. All tests were performed in triplicate and average values were calculated to ensure experimental accuracy.



**Figure 1.** Viscosity-temperature relationship of heavy oil. (a) Viscosity ( $\eta$ ) as a function of temperature ( $T$ ) in semilogarithmic coordinates. (b) The relationship between  $\ln\eta$  and temperature ( $T^{-1}$ ) in linear coordinates.

**2.3. Thixotropy of Heavy Oil.** The sample to be tested was placed in the high-temperature high-pressure rheometer. The target pressure was set, and the system was preheated to the target temperature and held for 1 h to ensure sufficient thermal equilibrium. The shear rate range was set from 0 to  $120 \text{ s}^{-1}$ . The testing procedure involved linearly increasing the shear rate from  $0 \text{ s}^{-1}$  to  $120 \text{ s}^{-1}$  over 20 min, followed by immediately decreasing it back to  $0 \text{ s}^{-1}$  over another 20 min at a uniform rate. All tests were performed in triplicate, and average values were calculated to ensure experimental accuracy.

**2.4. Impact of  $\text{CO}_2$  on Crude Oil Rheology.**  
**2.4.1. Experiment on the Solubility of  $\text{CO}_2$ .** The pretreated J230 heavy oil sample ( $50.0 \pm 0.1 \text{ g}$ ) was injected into a high-pressure cylinder using a high-pressure saturation device. After vacuum degassing,  $\text{CO}_2$  was injected into the system. The mixture was presaturated for 24 h under target pressure and temperature to ensure gas–liquid phase equilibrium. This was followed by a constant-volume depletion test, during which the volume of liberated gas was measured using a wet gas flow meter to calculate the dissolved gas-oil ratio. Throughout the process, the pressure was controlled with an accuracy of  $\pm 0.05 \text{ MPa}$ , the temperature fluctuation was within  $\pm 0.1^\circ \text{C}$ , and the volume measurement error did not exceed  $\pm 0.5\%$ . Triplicate parallel experiments were conducted to ensure reproducibility.

**2.4.2. The Rheological Properties of Heavy Oil after Dissolving  $\text{CO}_2$ .** During the development of heavy oil utilizing  $\text{CO}_2$ -thermal synergistic effects, the process is subject to the combined influence of thermal effects and formation shear. To clarify the impact patterns of thermal history and shear history on the overall development process of heavy oil under different working conditions, characteristic temperature and pressure points were selected based on solubility experimental results for rheological testing of  $\text{CO}_2$ -saturated heavy oil. A high-pressure rheometer was employed to conduct viscosity-temperature, thixotropic hysteresis loop, and dynamic strain tests, following the procedures described in Sections 2.2 and 2.3. The study focused on quantifying thermal history effects (assessing irreversible damage through heating–cooling cycles) and shear history effects (structural reconstruction of heavy oil due to stepwise shear), simulating the scenarios of injection-production disturbance and thermal recovery. All experiments were performed in triplicate under each condition, with key parameters controlled within specified coefficients of variation to ensure data repeatability.

### 3. RESULT AND DISCUSSION

**3.1. Modified Arrhenius Model and Viscosity Chart for Heavy Oil.**  
**3.1.1. Viscosity-Temperature Characteristics of Heavy Oil.** Viscosity is a key parameter characterizing the flow capacity of heavy oil. The high viscosity of heavy oil leads to significant flow resistance, severely restricting its flowability within the reservoir. Therefore, reducing viscosity constitutes a core challenge in heavy oil development.<sup>30,31</sup> This complexity originates from the three-dimensional network structure formed by asphaltenes and resins,<sup>32</sup> whose topological properties are highly sensitive to the temperature field. When temperature increases, molecular thermal motion disrupts the connecting points within the colloidal network, resulting in structural disaggregation and a sharp decline in viscosity.<sup>33,34</sup> Traditionally, this temperature dependence is commonly described by the Arrhenius equation,<sup>35–37</sup> as shown in Equation 1a.

$$\eta = Ae^{E_a/RT} \quad (1a)$$

$$\ln\eta = \ln A + \frac{E_a}{R} \cdot \frac{1}{T} \quad (1b)$$

where  $\eta$  is the viscosity in  $\text{mPa}\cdot\text{s}$ ,  $E_a$  is the activation energy in  $\text{J/mol}$ ,  $A$  is the pre-exponential factor,  $R$  is the universal gas constant equal to  $8.314 \text{ J}/(\text{mol}\cdot\text{K})$ , and  $T$  is the absolute temperature in  $\text{K}$ .

However, the Arrhenius model is only applicable for predicting the viscosity of heavy oil at high temperature. At low temperature, the viscosity of heavy oil is no longer a single function of temperature.<sup>38</sup> When the temperature falls below a certain threshold, the spatial structure of heavy oil undergoes significant alterations: resin aggregates increase substantially, structural complexity rises abruptly, and a three-segment linear pattern emerges in logarithmic coordinates. Distinct temperature transition points exist. Above these points, heavy oil gradually transitions from non-Darcy flow to Darcy flow, and its temperature sensitivity progressively weakens, resulting in a flatter viscosity-temperature curve in logarithmic coordinates. As the temperature continues to increase, the spatial structure of resin aggregates in heavy oil is further disrupted. Beyond a specific threshold, the flow behavior shifts entirely to Darcy flow, becoming solely a function of temperature. In Figure 1, these three regions are defined as the non-Newtonian flow zone, the quasi-Darcy flow zone, and the Darcy flow zone, respectively.



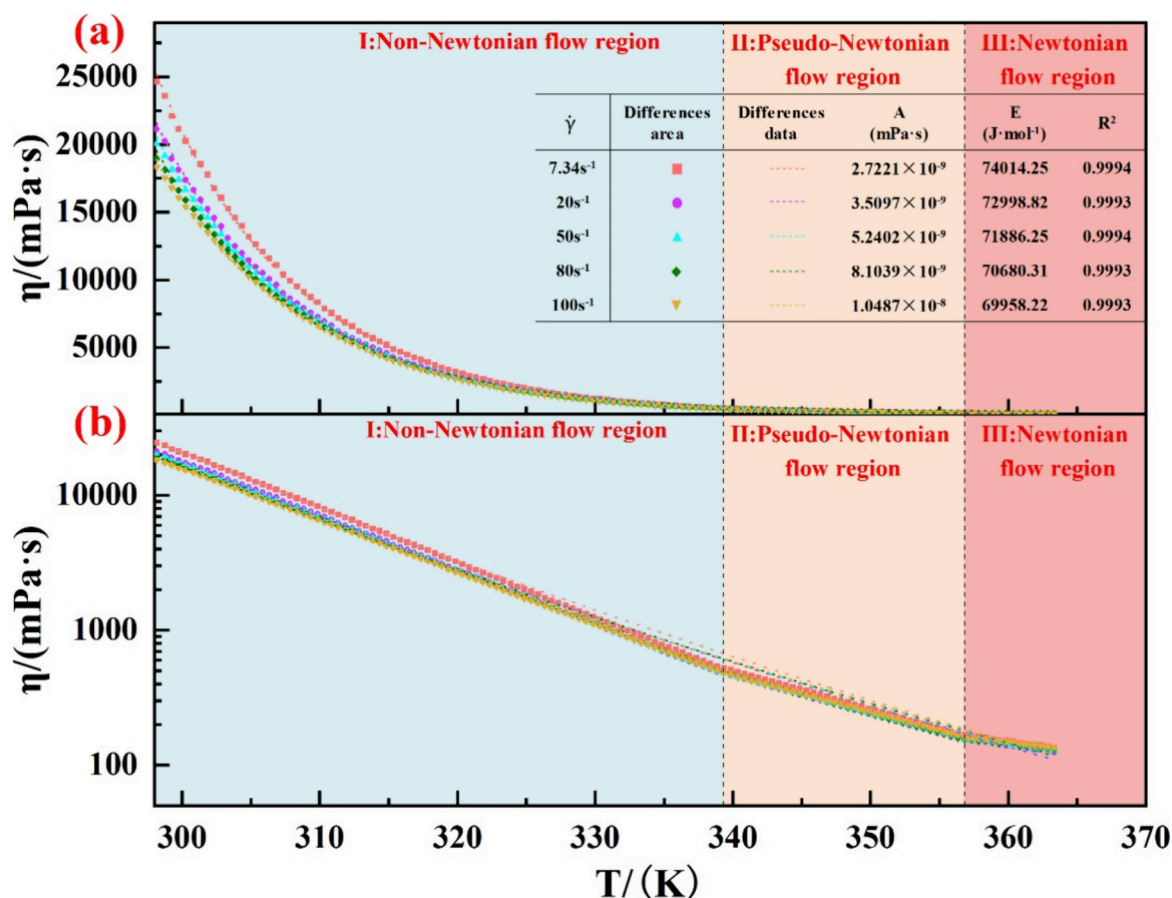


Figure 2. Viscosity-temperature relationship prediction model. (a) Form of  $\eta$  -  $T$ , (b) Form of  $\ln\eta$  -  $T$ .

The three distinct regions were individually fitted according to the Arrhenius equation, yielding the respective formulas for each zone as shown in eq 1b. In the low-temperature region ( $T < 339.35$  K), the activation energy is the highest (79.13 kJ/mol), indicating that the resin aggregate structure remains intact and molecular motion must overcome a high energy barrier. When the temperature rises into the transition region ( $339.35 \text{ K} \leq T \leq 357.35 \text{ K}$ ), the activation energy decreases to 66.4 kJ/mol. The reduction in slope reflects the onset of resin aggregate disaggregation, leading to a transition from non-Darcy to Darcy flow. In the high-temperature region ( $T > 357.35 \text{ K}$ ), the activation energy drops sharply to 34.84 kJ/mol. At this stage, resin aggregates are thoroughly disrupted, and the system transitions into a Newtonian fluid. It is worth noting that the pre-exponential factor  $A$  continuously increases with rising temperature, reflecting enhanced molecular transport frequency and effective collision probability. Together with the decrease in activation energy, this trend reveals the fundamental mechanism by which thermal agitation reduces the flow barrier by weakening intermolecular forces. The two characteristic temperature inflection points correspond to critical transition thresholds in the rheological behavior of the heavy oil.

$$\eta = \begin{cases} 2.8848 \times 10^{-9} e^{73866.84/RT}, R^2 = 0.9993, \text{ I;} \\ 3.9896 \times 10^{-8} e^{65768.64/RT}, R^2 = 0.9997, \text{ II;} \\ 0.00119 e^{35158.6/RT}, R^2 = 0.9961, \text{ III.} \end{cases} \quad (2)$$

**3.1.2. The Modified Arrhenius Equation for Heavy Oil.** As evidenced by the preceding analysis, a significant deviation exists between the measured viscosity of heavy oil and the viscosity predicted by the Arrhenius equation, primarily stemming from the high sensitivity of its internal structure to temperature. To elucidate the true viscosity-temperature response of heavy oil under reservoir conditions, this study systematically conducted viscosity-temperature tests at five different shear rates ( $\dot{\gamma}$ ). Figure 2 presents the viscosity-temperature curves of the heavy oil under these varying shear rates.

It can be observed that at high shear rates, the apparent viscosity of the heavy oil further decreases, while still exhibiting a three-stage linear pattern in logarithmic coordinates. This reflects a structural transformation in the rheological behavior of heavy oil under the coupled effect of temperature and shear. These results indicate that the structure of heavy oil is dually influenced by both shear and temperature, with viscosity being significantly affected by shear rate in the non-Newtonian flow region at low temperatures. The traditional Arrhenius model, which assumes a linear relationship between  $\ln\eta$  and  $T^{-1}$ , does not account for the effect of shear, leading to insufficient predictive accuracy. Specifically, in the low-temperature region, shear forces have a strong disruptive effect on the heavy oil's structure, resulting in dramatic viscosity changes. In contrast, in the high-temperature region, where thermal agitation dominates the weakening of intermolecular forces, the influence of shear is relatively reduced. This leads to a distinct segmented linear characteristic in the viscosity-temperature



relationship across different shear rates when plotted in logarithmic coordinates.

Both the activation energy  $E_a$  and the pre-exponential factor  $A$  exhibit functional relationships with the shear rate  $\dot{\gamma}$ . Based on this, the Arrhenius equation model was applied to fit the viscosity-temperature data obtained at different shear rates, resulting in fitted models (eq 2) for each shear rate condition.

$$\eta = \begin{cases} 2.7221 \times 10^{-9} e^{74014.25/RT}, R^2 = 0.9994, 7.34s^{-1}; \\ 3.5097 \times 10^{-9} e^{72998.82/RT}, R^2 = 0.9993, 20s^{-1}; \\ 5.2402 \times 10^{-9} e^{71886.25/RT}, R^2 = 0.9994, 50s^{-1}; \\ 8.1039 \times 10^{-9} e^{70680.31/RT}, R^2 = 0.9993, 80s^{-1}; \\ 1.0487 \times 10^{-8} e^{69958.22/RT}, R^2 = 0.9993, 100s^{-1}; \end{cases} \quad (3)$$

To further quantify the influence of shear rate ( $\dot{\gamma}$ ) on the viscosity-temperature relationship of heavy oil, secondary fitting was performed on the Arrhenius fitting parameters—specifically the pre-exponential factor  $A$  and the activation energy  $E_a$ —obtained at different shear rates (Figure 3). The

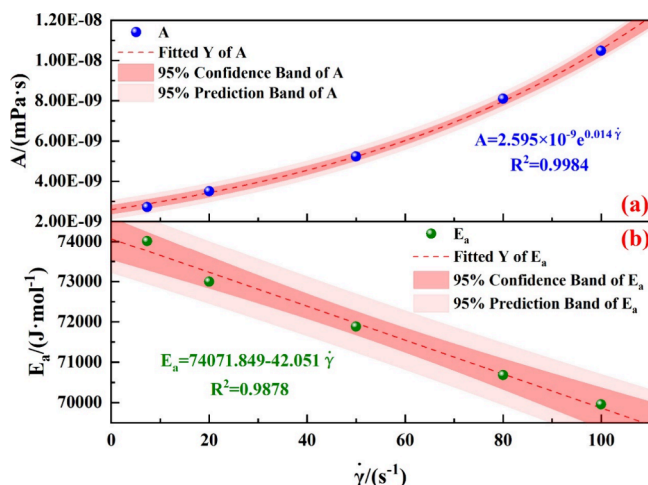


Figure 3. (a) Pre-exponential factor, (b) pseudoactivation energy.

results indicate that  $A$  increases exponentially with  $\dot{\gamma}$  (as shown in eq 3), while  $E_a$  decreases linearly (eq 4). The goodness-of-fit for both relationships exceed 0.98, further validating the significant regulatory effect of shear on the rheological behavior of heavy oil. This observed pattern demonstrates that an increase in shear rate not only promotes the disentanglement of molecular chains but also enhances the dynamic intermolecular response, collectively leading to a nonlinear reduction in the viscosity of heavy oil

$$A = 2.595 \times 10^{-9} e^{0.01404\dot{\gamma}}, R^2 = 0.9984 \quad (4)$$

$$E_a = 74071.849 - 42.051\dot{\gamma}, R^2 = 0.9978 \quad (5)$$

Based on the above analysis, the equations describing the relationships between the pre-exponential factor  $A$  (eq 3) and activation energy  $E_a$  (eq 4) with shear rate were incorporated into the Arrhenius equation (eq 1a), ultimately establishing a viscosity prediction model for heavy oil that simultaneously accounts for both temperature and shear rate effects (eq 5). This model overcomes the limitation of the traditional Arrhenius equation, which considers only the temperature

effect, and achieves for the first time an accurate characterization of heavy oil viscosity under the coupled influence of temperature and shear rate. To intuitively illustrate the variation of viscosity with temperature and shear rate, a three-dimensional viscosity distribution plot was further developed (Figure 4). This plot enables rapid determination

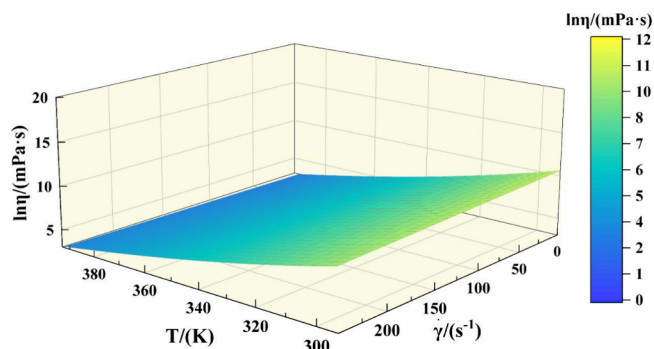


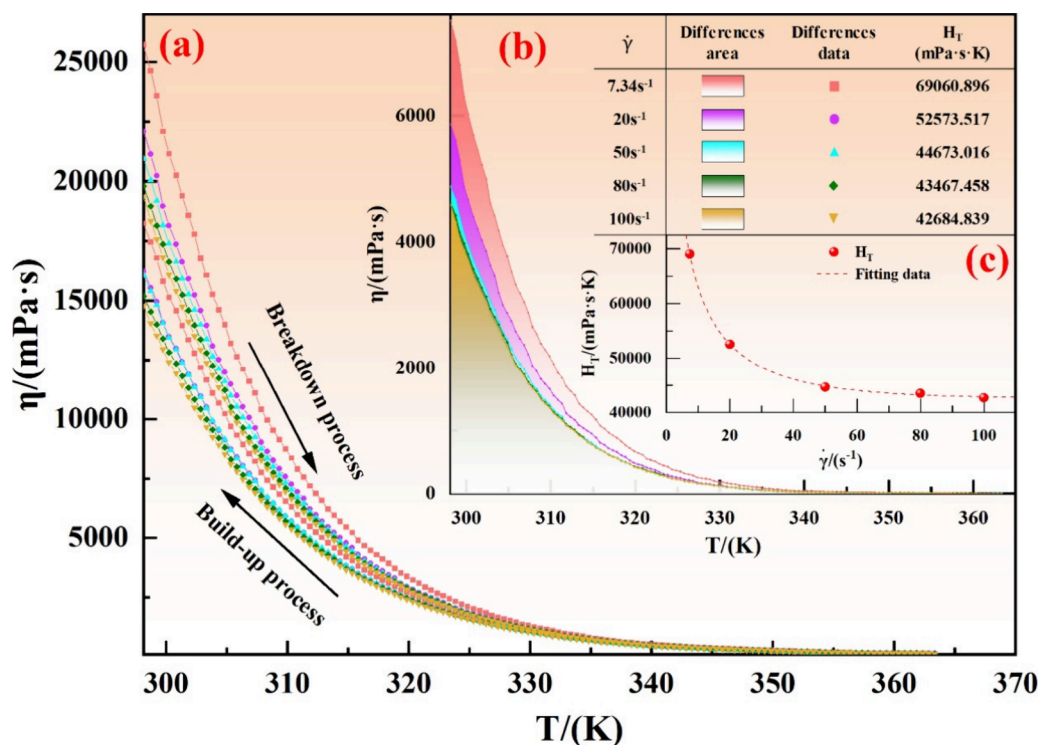
Figure 4. Viscosity of heavy oil at different temperature and shear rates.

of the heavy oil viscosity value for any temperature–shear rate combination, providing an important theoretical basis and engineering guidance for the optimization of thermal recovery processes parameters in oilfields.

$$\ln \eta = \ln(2.5946 \times 10^{-9}) + 0.01404\dot{\gamma} + \frac{74071.849 - 42.051\dot{\gamma}}{RT} \quad (6)$$

**3.2. The Thixotropy of Heavy Oil.** The previous sections systematically examined the variation of heavy oil viscosity with temperature and shear rate, demonstrating that thermal treatment is a key viscosity-reduction technique in heavy oil development. However, under actual reservoir conditions, crude oil undergoes a complete temperature cycle from heating to cooling. Studying only the heating phase is insufficient for a comprehensive evaluation of thermal recovery performance. It is therefore necessary to conduct a systematic assessment of the entire heating and cooling processes experienced by heavy oil during production, in order to reveal the impact of thermal cycling on its rheological behavior and the underlying structural response mechanisms. Heavy oil exhibits a viscosity-temperature hysteresis phenomenon during heating–cooling cycles, which fundamentally arises from the alteration of the asphaltene-resin supramolecular network by thermal history.<sup>39,40</sup> When the temperature crosses a critical transition threshold, thermal agitation overcomes the energy barriers of van der Waals forces and hydrogen bonds, leading to the dissociation of the three-dimensional network. This causes asphaltene layers to melt into discrete units and resin molecules to desorb from asphaltene surfaces. During the subsequent cooling processes, structural reorganization is dominated by kinetic nonequilibrium recombination. The dispersed asphaltenes act as nucleation sites triggering disordered stacking, while resins cannot fully restore their original coverage due to irreversible loss of adsorption sites. The synergistic effect of these processes results in a new, loose and porous structure, in which steric hindrance significantly inhibits the reformation of a dense network.

This structural transformation causes the viscosity during cooling to be systematically lower than that during heating at



**Figure 5.** Influence of heat treatment on the viscosity of heavy oil at different shear rates. (a) Viscosity–temperature relationship, (b) Hysteresis loop area, (c) Form of  $H_T \cdot \dot{\gamma}$ .

the same temperature. To quantify the degree of structural alteration, the hysteresis loop area  $H_T$  is introduced as a key indicator. It is defined as the enclosed area between the apparent viscosity ( $\eta$ ) versus temperature ( $T$ ) curves during the heating and cooling cycles, as given by the formula:

$$H_T = \int_0^T \eta(T) dT \quad (7)$$

where  $H_T$  is the viscosity-temperature hysteresis area (mPa·s·°C),  $\eta$  is the viscosity (mPa·s), and  $T$  is the temperature (°C). The value of  $H_T$  quantitatively represents the structural recovery capability of heavy oil before and after thermal treatment.

Figure 5 illustrates the effects of thermal treatment on heavy oil under different shear rates. Figure 5(b) shows the variation of hysteresis area with temperature. Experimental results confirm that the hysteresis area systematically decreases as temperature increases, indicating that high-temperature conditions weaken the self-repair capacity of the heavy oil network. Furthermore, the hysteresis area also decreases with increasing shear rate, demonstrating that shear similarly influences the structure. At high shear rates, the effect of temperature on the hysteresis behavior is reduced. To visually represent the relationship between hysteresis area and shear rate, Figure 5(c) is provided. The hysteresis area serves as a key criterion for optimizing thermal recovery processes. A larger hysteresis area suggests difficult recombination of the asphaltene-resin network, necessitating significantly extended shut-in time to ensure production capacity recovery. This parameter also guides the dynamic adjustment of steam injection rates to compensate for viscosity loss due to structural damage. Combined with the previously established viscosity prediction model, it enables accurate prediction of load fluctuations in lifting equipment under variable-speed conditions. This

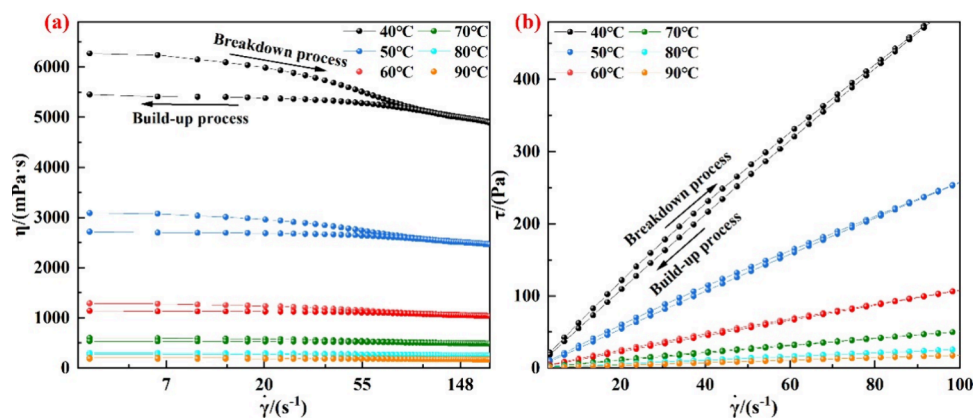
method establishes a bridge between molecular-scale structural characteristics and reservoir engineering decisions through rheological parameters, providing a theoretical tool for the efficient development of heavy oil.

As demonstrated in the preceding sections, the rheological behavior of heavy oil is influenced not only by thermal effects but also significantly by shear, exhibiting a pronounced shear-thinning phenomenon where the apparent viscosity decreases with increasing shear rate. The microscopic origin of this behavior lies in the three-dimensional network structure formed by resin and asphaltene molecules through intermolecular forces such as van der Waals interactions and hydrogen bonding.<sup>41,42</sup> When external shear is applied, weak bonds between molecules are selectively broken, resin chains disentangle to reduce flow resistance, sheet-like asphaltenes align along the flow direction to minimize turbulent dissipation, and aggregate breakdown leads to a notable reduction in micelle size. This series of structural changes results in a characteristic power-law decay in viscosity. While the breakdown of the network occurs over short time scales, structural reorganization requires a much longer relaxation time ranging from minutes to hours. This significant difference in time scales forms the kinetic basis for the observed thixotropic behavior.

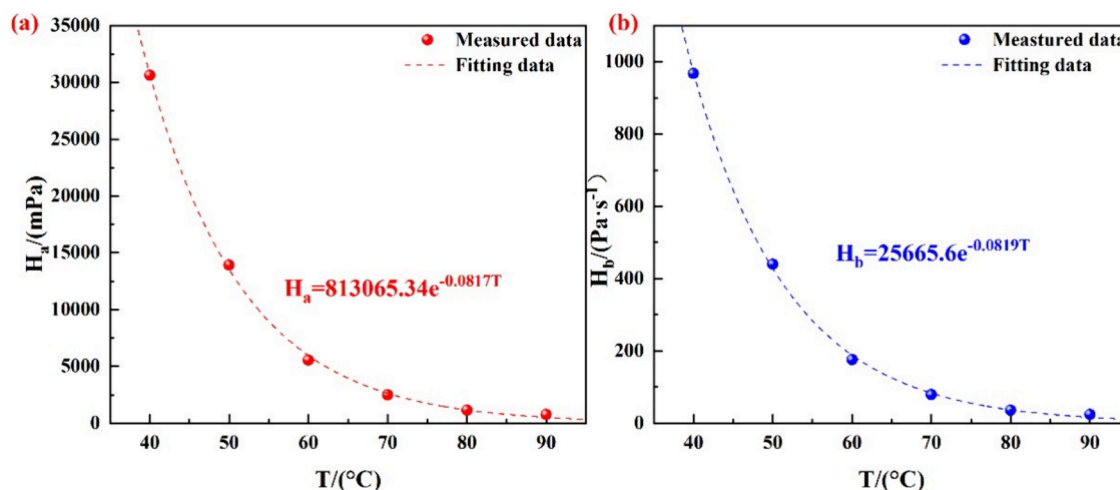
To quantitatively analyze the influence of shear history on the thixotropic behavior of heavy oil, this study defines two key rheological parameters: the hysteresis area of shear viscosity  $H_a$  and the hysteresis area of shear force  $H_b$  (eq 7).

$$H_a = \int_0^{\dot{\gamma}} \eta(\dot{\gamma}) d\dot{\gamma} \quad (8a)$$

$$H_b = \int_0^{\dot{\gamma}} \tau(\dot{\gamma}) d\dot{\gamma} \quad (8b)$$



**Figure 6.** Influence of shear treatment on the viscosity of heavy oil at different temperature. (a) Form of  $\eta - \dot{\gamma}$ , (b) Form of  $\tau - \dot{\gamma}$ .



**Figure 7.** Shear treatment loop area as a function of temperature. (a) Form of  $H_a - T$ , (b) Form of  $H_b - T$ .

where  $H_a$  represents the irreversible loss of viscosity during the shear rate ascending/descending cycle (mPa), and  $H_b$  denotes the net energy consumption of structural destruction ( $\text{Pa} \cdot \text{s}^{-1}$ ). Here,  $\eta$  is the viscosity ( $\text{mPa} \cdot \text{s}$ ),  $\tau$  is the shear stress (Pa), and  $\dot{\gamma}$  is the shear rate ( $\text{s}^{-1}$ ).

Together, these two parameters reveal the essential mechanism of thixotropy in heavy oil. During the shear rate ascending phase, heavy oil exhibits significant viscosity attenuation. Shear forces destroy the resin-asphaltene network, leading to structural dissociation and a reduction in macromolecular entanglement density, which causes a pronounced decrease in viscosity. In the descending phase, however, due to the fact that the reorganization relaxation time is much longer than the destruction time scale, the reorganization kinetics are delayed, preventing the viscosity from recovering to its initial value and resulting in the formation of a characteristic hysteresis loop. As temperature increases, both  $H_a$  and  $H_b$  show systematic attenuation (Figure 6, the unit of temperature is  $^{\circ}\text{C}$ ). When the temperature crosses the anomalous transition threshold, the hysteresis area decreases sharply, indicating a loss of self-repair capacity in the three-dimensional network.

Figure 7 reveals that the hysteresis area of shear viscosity  $H_a$  and the hysteresis area of shear force  $H_b$  evolve in three distinct stages as temperature increases. In the low-temperature region ( $T < 80^{\circ}\text{C}$ ), both  $H_a$  and  $H_b$  remain at high levels, indicating significant thixotropic behavior. When the temperature crosses the anomalous transition threshold, the hysteresis

areas decrease sharply, marking a sudden loss of self-repair capacity in the resin-asphaltene network. In the high-temperature region,  $H_a$  and  $H_b$  approach zero, confirming the transition of heavy oil to Newtonian fluid behavior under high-temperature conditions. This phenomenon demonstrates the dual regulatory effect of temperature on shear-thinning behavior: at low temperature, it enhances the hysteresis of structural reorganization, while at high temperature, network dissociation leads to the disappearance of thixotropy. This method can serve as an objective criterion for identifying rheological anomaly points, as  $H_a$  quantitatively reflects the energy consumption of structural destruction, and  $H_b$  characterizes the intensity of reorganization kinetic hysteresis. This phenomenon holds significant engineering guidance value. During the steam injection phase, the temperature should exceed the critical transition point to achieve sufficient dissociation of the resin-asphaltene network structure. When designing the production strategy, the shut-in time should be scientifically determined based on the exponential decay law of thixotropic strength to balance productivity recovery and production efficiency. For production performance prediction, the degree of structural damage should be incorporated into constitutive equations and numerical models to enhance the accuracy of production forecasts. This understanding not only clarifies the rheological control mechanism of the  $\text{CO}_2$ -thermal synergistic effect but also provides direct theoretical support for optimizing injection-production parameters and predicting



development performance in field applications, effectively bridging fundamental research and engineering practice.

**3.3. The Influence of CO<sub>2</sub> on the Viscosity of Heavy Oil.** The solubility of CO<sub>2</sub> in heavy oil ( $S$ , m<sup>3</sup>/m<sup>3</sup>) exhibits a significant temperature–pressure coupling effect. Figure 8

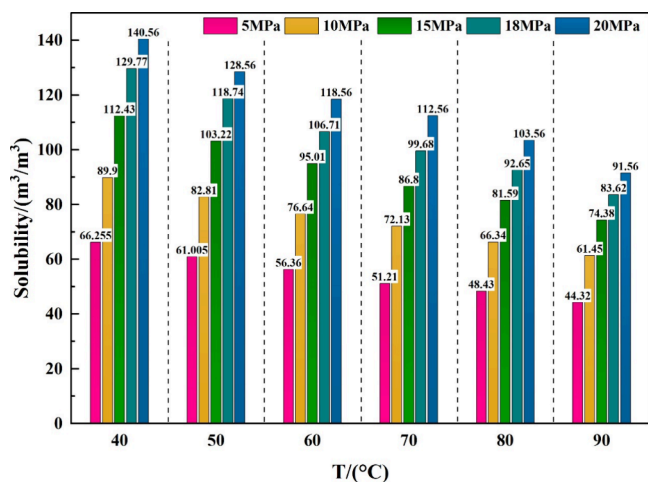


Figure 8. Solubility of CO<sub>2</sub> at different temperature and pressures.

illustrates the variation of CO<sub>2</sub> solubility under different temperature and pressure conditions. Driven by pressure, CO<sub>2</sub> molecules diffuse into the interstitial spaces of the resin-asphaltene network, expanding the free volume of the system and increasing the molecular spacing. Simultaneously, CO<sub>2</sub> forms weak bonding structures with alkane chains through van der Waals forces, an interaction that attenuates with increasing temperature, leading to reduced solubility.<sup>43–47</sup> The synergistic effect of temperature and pressure shows a linear response, while temperature rise weakens the van der Waals forces between CO<sub>2</sub> and alkane chains, promoting desorption. This negative temperature effect causes  $S$  to decrease with increasing temperature; however, more intense molecular motion at elevated temperature shortens the time required to achieve diffusion equilibrium.

Under simulated target reservoir pressure conditions (3 MPa), systematic measurements of the viscosity reduction rate ( $\theta$ , defined as the percentage decrease in crude oil viscosity, %) of CO<sub>2</sub>-saturated heavy oil were conducted at different shear rates (7.34, 20, 50, 75, and 100 s<sup>−1</sup>) and temperature (40, 50, 70, and 90 °C) (Figure 9). The results reveal a significant shear-temperature coupling effect on the viscosity-reduction efficiency of CO<sub>2</sub>. As the shear rate increases, the viscosity reduction rate systematically rises. In the high-shear region ( $\dot{\gamma}$  = 100 s<sup>−1</sup>), the viscosity loss reaches 65.1%, which is 3.3% higher than that at the low shear rate ( $\dot{\gamma}$  = 7.34 s<sup>−1</sup>) of 61.8%. In contrast, at a constant shear rate, the viscosity reduction rate follows an exponential decay with increasing temperature, decreasing by 2.8% across the temperature range from 40 to 90 °C. This phenomenon originates from the microscopic interaction mechanisms within the CO<sub>2</sub>-heavy oil system. Dissolved CO<sub>2</sub> expands the free volume and increases molecular spacing, thereby weakening van der Waals forces. However, elevated temperature intensifies molecular thermal motion, leading to the attenuation of weak bonding interactions between CO<sub>2</sub> and alkane chains, as well as a decrease in solubility. Simultaneously, high shear rates disrupt swelling stability, induce local degassing and phase separation,

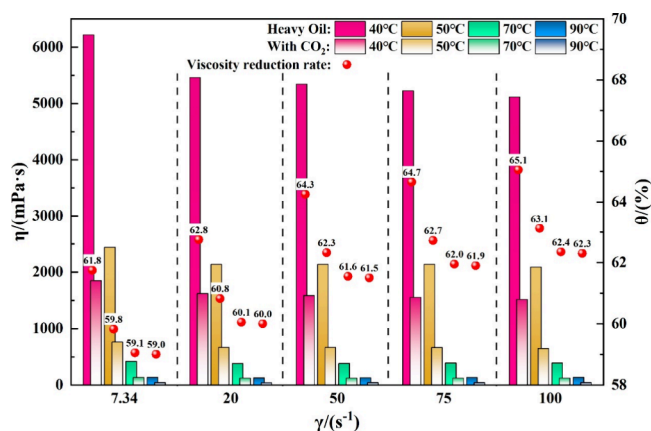


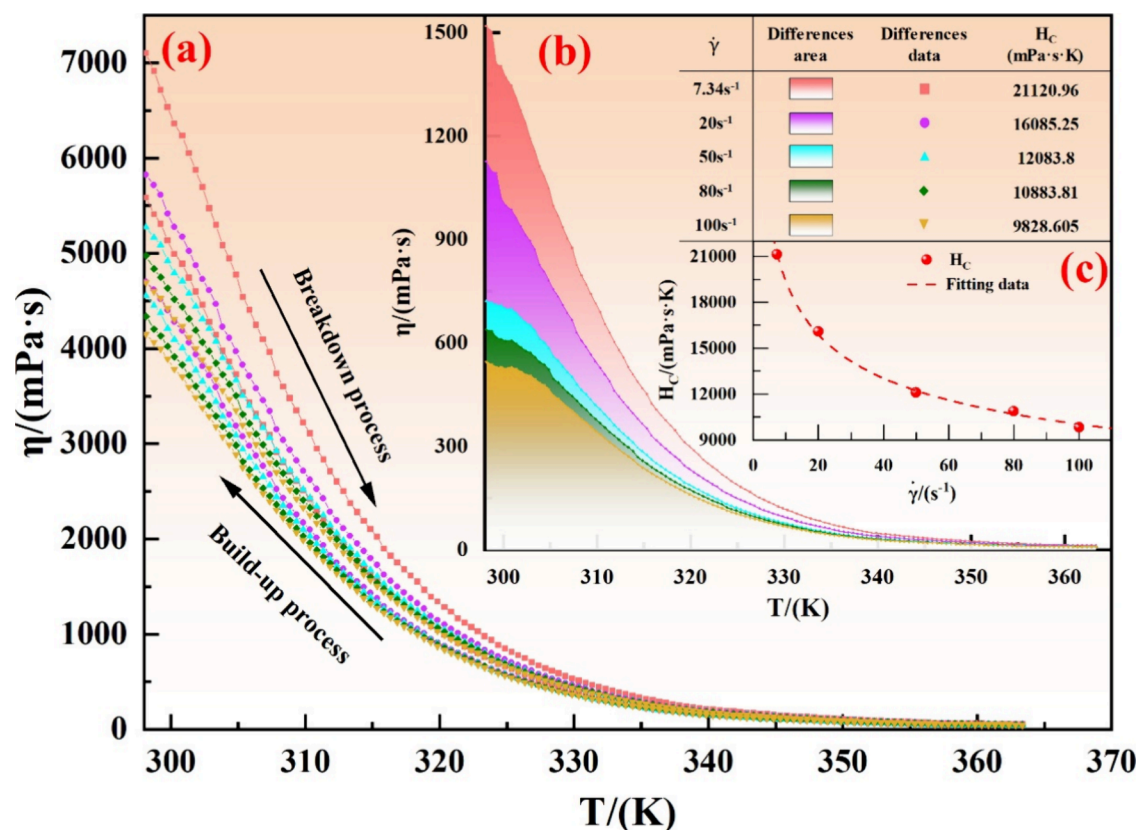
Figure 9. Variation of the viscosity reduction rate of CO<sub>2</sub> at different temperature and shear rates.

and reduce the effective dissolved concentration. The regulatory effect of temperature on the state of the resin-asphaltene network further amplifies this behavior: in the low-temperature region, the resin aggregate structure remains intact, enhancing CO<sub>2</sub> disruptive effect on the heavy oil structure; in the high-temperature region, although the network disaggregates, thermal disturbance dominates viscosity reduction, causing the marginal contribution of CO<sub>2</sub> to diminish sharply. In summary, the synergistic effect of temperature and shear collectively influences the viscosity-reduction performance of CO<sub>2</sub> in heavy oil. These findings suggest that near-wellbore regions require steam coinjection to compensate for heat loss or a reduction in gas injection rate. The critical inflection point predicted by the exponential decay model can be used to optimize shut-in time. For shallow low-temperature reservoirs, a "CO<sub>2</sub>-thermal" hybrid system is recommended, employing high-rate injection to enhance the viscosity-reduction effect. This study provides molecular-scale mechanistic support and engineering optimization pathways for multithermal fluid displacement technologies.

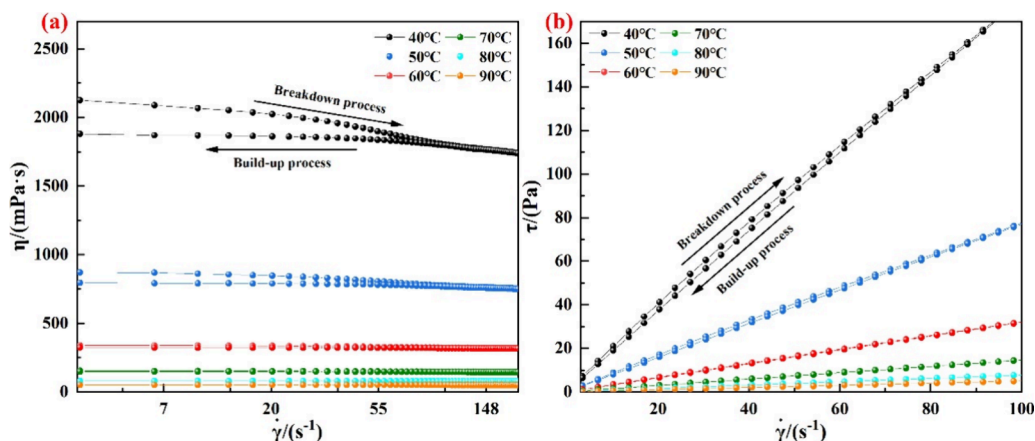
To further quantify the impact of thermal treatment on CO<sub>2</sub>-saturated heavy oil, shear tests were conducted at different shear rates, as shown in Figure 10. Thermal treatment influences the CO<sub>2</sub>-dissolved heavy oil in a manner similar to its effect on deaerated heavy oil, but the absolute value of the thixotropic hysteresis loop area is significantly reduced (Figure 10c). The hysteresis loop area exhibits an exponential decay with increasing shear rate.

Through systematic testing of the rheological behavior of thermally treated CO<sub>2</sub>-saturated heavy oil under different shear rates (7.34, 20, 50, 75, and 100 s<sup>−1</sup>) (Figure 10), it was observed that thermal treatment significantly reduces the hysteresis loop area  $H_c$  of the CO<sub>2</sub>-saturated oil (Figure 10c). Taking the shear rate of 7.34 s<sup>−1</sup> as an example, the  $H_c$  value decreased from 69,060.896 for the original heavy oil to 21,120.96 for the CO<sub>2</sub>-saturated oil after thermal treatment, representing a reduction of 69.42%.

Furthermore,  $H_c$  follows an exponential decay trend as the shear rate increases. This phenomenon stems from the synergistic damaging effects of thermal treatment and CO<sub>2</sub> dissolution. High temperature promote the desorption of resins from asphaltene surfaces, forming discrete units and a loose porous structure. After CO<sub>2</sub> dissolution, the kinetic energy of the heavy oil surpasses the recombination activation energy, weakening the recombination capability of heavy oil



**Figure 10.** Influence of shear treatment on the viscosity of heavy oil saturated with CO<sub>2</sub> at different shear rates. (a) Viscosity–temperature relationship, (b) Hysteresis loop area, (c) Form of  $H_c - \dot{\gamma}$ .

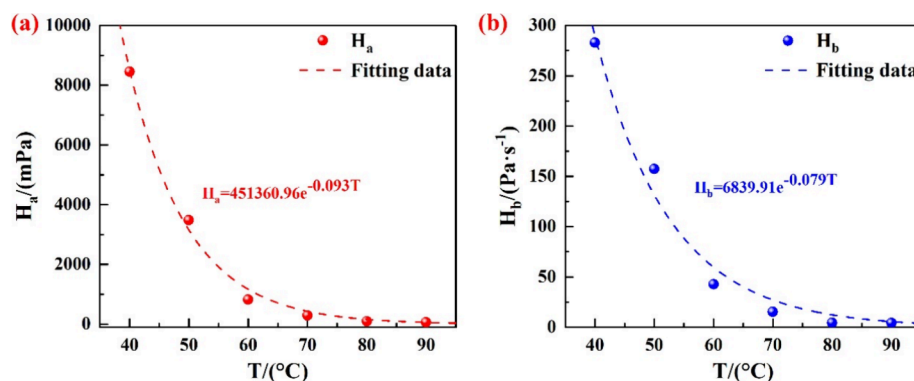


**Figure 11.** Influence of share treatment on the viscosity of heavy oil saturated with CO<sub>2</sub> at different temperature. (a) Form of  $\eta - \dot{\gamma}$ , (b) Form of  $\tau - \dot{\gamma}$ .

macromolecules, inhibiting the reassociation of resins and asphaltenes, and leading to more thorough molecular disentanglement, thereby reducing the hysteresis effect. This effect indicates that although thermal treatment can reduce thixotropic energy consumption, the viscosity of the heavy oil remains relatively high after structural recovery, which may still cause near-wellbore clogging. It is necessary to dynamically adjust the shut-in time based on the decay model to fully leverage the synergistic effects of CO<sub>2</sub> and heat in enhancing the fluidity of heavy oil. This study reveals that thermal treatment weakens thixotropy through structural modification, but the shear dependence of the recombination energy barrier

necessitates a process design that synergistically incorporates "thermal-rheological-chemical" multifield effects.

**3.4. Viscoelasticity of Saturated CO<sub>2</sub> Heavy Oil and Constitutive Equation.** To investigate the temperature–shear coupling effects of CO<sub>2</sub> dissolution on the rheological behavior of heavy oil, systematic shear tests were conducted within the temperature range of 40–90 °C on CO<sub>2</sub>-saturated heavy oil (Figure 11). This study reveals a profound restructuring mechanism induced by CO<sub>2</sub> dissolution: both the viscosity hysteresis area ( $H_a$ ) and the shear stress hysteresis area ( $H_b$ ) decreased sharply compared to those of the original heavy oil, indicating that CO<sub>2</sub> penetration disrupts hydrogen bonding and  $\pi$ - $\pi$  stacking within the resin layered structure,



**Figure 12.** Shear treatment loop area as a function of temperature. (a) Form of  $H_a - T$ , (b) Form of  $H_b - T$ .

significantly impairing the recovery capacity of the resin-asphaltene network.<sup>48–51</sup> The synergistic temperature–thixotropy evolution (Figure 12) further demonstrates that at temperature above 70 °C, the values of  $H_a$  and  $H_b$  correspond to effects equivalent to those observed in non- $\text{CO}_2$ -solubilized systems above 80 °C—effectively reducing the threshold temperature by 10 °C. This phenomenon results from intensified  $\text{CO}_2$  diffusion at elevated temperature, where  $\text{CO}_2$  molecules penetrate micellar interstices, weakening van der Waals forces and polar interactions, and disentangling the macromolecular system from a highly entangled state to a low-entanglement regime. Concurrently,  $\text{CO}_2$  extracts light components and reduces interfacial tension, synergizing with thermal motion to diminish network strength. As a result, the heavy oil exhibits near-Newtonian behavior at 70 °C, representing a significant efficiency improvement over thermal treatment alone. This mechanism effectively lowers the operational temperature threshold for extraction to 70 °C and reduces steam injection volume by 15–20%. It demonstrates that in low-temperature regions, the synergistic action of  $\text{CO}_2$  and heat can enhance the flowability of heavy oil, providing a molecular-scale design basis for multithermal fluid stimulation technologies.

The shear stress versus shear rate curve during the heating processes in Figure 11(b) was fitted and found to conform to the Bingham model (eq 8a).

$$\tau = \tau_0 + K\dot{\gamma} \quad (9)$$

In the equation,  $\tau$  is the shear stress, Pa;  $\tau_0$  is the yield stress, Pa;  $\dot{\gamma}$  is the shear rate,  $\text{s}^{-1}$ , and  $K$  is the consistency coefficient,  $\text{Pa}\cdot\text{s}$ .

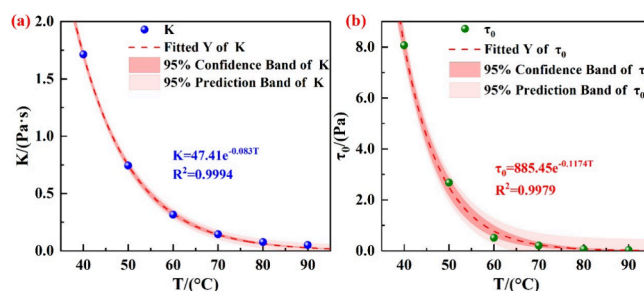
The fitting results are summarized in Table 2. Even after  $\text{CO}_2$  dissolution, the heavy oil continues to exhibit yield characteristics dominated by the resin-asphaltene network skeleton, retaining a measurable yield stress. The fitted line does not pass through the origin, confirming that the

**Table 2.** Rheological Curve Constitutive Equation and Yield Stress

| Crude Oil | $T$ /(°C) | $T$ /(Pa)                        | $R^2$ |
|-----------|-----------|----------------------------------|-------|
| Heavy oil | 40        | $\tau = 1.71\dot{\gamma} + 8.07$ | 99.96 |
|           | 50        | $\tau = 0.74\dot{\gamma} + 2.68$ | 99.97 |
|           | 60        | $\tau = 0.32\dot{\gamma} + 0.51$ | 99.96 |
|           | 70        | $\tau = 0.15\dot{\gamma} + 0.21$ | 99.99 |
|           | 80        | $\tau = 0.08\dot{\gamma} + 0.07$ | 99.99 |
|           | 90        | $\tau = 0.05\dot{\gamma} + 0.03$ | 99.99 |

rheological behavior follows the Bingham model.<sup>52</sup> This indicates that flow initiation requires overcoming the yield stress to break the resin-asphaltene network structure.<sup>7,53</sup> As the temperature increases from 40 to 90 °C, the yield stress  $\tau_0$  decreases exponentially from 8.07 to 0.03 Pa (a reduction of 99.6%). Concurrently, the rheological curves shift systematically toward the horizontal axis with a decreasing slope, reflecting a continuous weakening of non-Newtonian behavior. In the low-temperature region (<60 °C), an increase in shear rate of  $50 \text{ s}^{-1}$  can reduce the apparent viscosity by up to 70%, whereas in the high-temperature region (>80 °C), a similar change in shear rate results in less than 5% viscosity variation. This evolution can be accurately described by the constructed temperature-shear universal constitutive equation.

The consistency coefficient  $K$  and the yield stress  $\tau_0$  from the constitutive equation in Table 2 were fitted against temperature, and their variations with temperature are shown in Figure 13. Both the consistency coefficient  $K$  and the yield



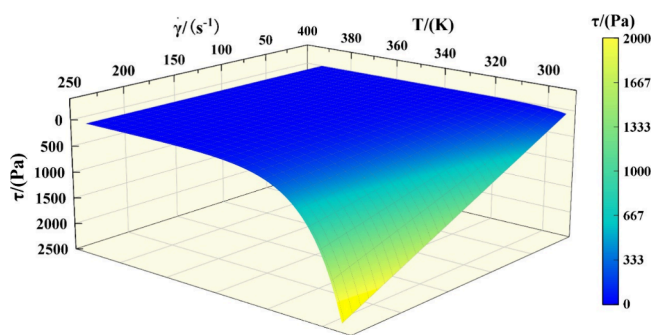
**Figure 13.** (a) Consistency coefficient factor, (b) Yield stress.

stress  $\tau_0$  in the constitutive equation of heavy oil exhibit a power-law exponential decay with increasing temperature. This behavior originates from the significant weakening of the resin-asphaltene network skeleton strength and the reduction in intermolecular viscous resistance caused by rising temperature. By substituting the fitted formulas into the Bingham model (eq 8a), a temperature-dependent universal constitutive equation for heavy oil was established (eq 8b). This equation couples the synergistic effects of temperature and shear rate on flow behavior, quantitatively characterizing the systematic evolution of rheological properties with temperature. It enables accurate prediction of the rheological properties of heavy oil across the entire temperature range.

$$\tau = 885.45e^{-0.1174T} + 47.41e^{-0.083T}\dot{\gamma} \quad (10)$$



Based on Equation 8b, a three-dimensional stress response surface for CO<sub>2</sub>-saturated heavy oil was plotted (Figure 14),



**Figure 14.** Shear stress of heavy oil at different temperature and shear rates.

visually revealing the evolution of shear stress ( $\tau$ ) under varying temperature and shear rate ( $\dot{\gamma}$ ) conditions. The stress exhibits an exponential decay with increasing temperature, while the influence of shear rate weakens significantly as temperature rises. In the low-temperature region, the surface shows a steep inclination, where a slight increase in  $\dot{\gamma}$  leads to a sharp decrease in stress, highlighting the strong shear-thinning behavior of the resin-asphaltene network. In contrast, the surface becomes nearly flat in the high-temperature region, where changes in  $\dot{\gamma}$  cause only minor stress adjustments, reflecting the near-Newtonian fluid behavior after the collapse of the colloidal network. This surface further identifies a sensitive transition zone between 50–60 °C where rheological properties change markedly, and an engineering safe zone above 70 °C where stress is dominated by viscous resistance, allowing for simplified operational design. The model provides a visual theoretical tool for optimizing production parameters in heavy oil extraction.

#### 4. CONCLUSION

This study establishes a novel methodology for quantifying thermal-shear history effects through hysteresis area analysis and develops a comprehensive temperature-dependent constitutive model that accurately predicts heavy oil rheological behavior across the full temperature range. These innovations provide both theoretical advances in understanding CO<sub>2</sub>-thermal synergistic mechanisms and practical tools for optimizing heavy oil recovery processes. The specific conclusions are as follows:

- (1) The phase transition critical conditions for the rheological behavior of heavy oil were identified. At temperature below 339.35 K, heavy oil exhibits non-Darcy flow characteristics with a high flow activation energy (79.13 kJ/mol). In the temperature range of 339.35–357.35 K, the activation energy decreases to 66.4 kJ/mol, the flow behavior transitions to Darcy flow. When the temperature exceeds 357.35 K, the activation energy further drops to 34.84 kJ/mol, resin aggregates completely disintegrate, and the system transitions to a Newtonian fluid.
- (2) The traditional viscosity-temperature prediction model for heavy oil was modified, and a new viscosity prediction model that simultaneously considers both temperature and shear rate was established. This model accurately characterizes the coupling effect of shear and

temperature fields under reservoir conditions, significantly improving prediction accuracy.

- (3) The viscosity reduction efficiency of CO<sub>2</sub> exhibits a notable temperature-shear coupling effect: the viscosity reduction rate increases with higher shear rates, reaching 65.1% at  $\dot{\gamma} = 100 \text{ s}^{-1}$ , which is 3.3% higher than at  $\dot{\gamma} = 7.34 \text{ s}^{-1}$ . At the same shear rate, the viscosity reduction rate decays exponentially with increasing temperature, decreasing by 2.8% over the 40–90 °C range.
- (4) The hysteresis area was introduced as a key indicator to quantify the impact of thermal-shear history on heavy oil structure. It was found that the hysteresis area decays exponentially with increasing shear rate and temperature. This parameter can serve as an important criterion for optimizing thermal recovery processes and, combined with the viscosity prediction model, accurately predict the load fluctuations of lifting equipment under variable-speed conditions.
- (5) The constitutive behavior of CO<sub>2</sub>-saturated heavy oil was determined to conform to the Bingham model, with both the consistency coefficient  $K$  and yield stress  $\tau_0$  following power-law decay with increasing temperature. Accordingly, a temperature-dependent universal constitutive equation was constructed. This model couples temperature and shear effects, enabling precise characterization of the rheological behavior of heavy oil across the entire temperature range, providing a theoretical basis for numerical simulation of oil reservoirs and the design of production processes.

This study provides a reliable theoretical basis and process optimization methods for the CO<sub>2</sub>-thermal synergistic development of conventional heavy oil. Extra-heavy and ultraheavy oils possess more complex spatial topological structures, leading to more intricate rheological response mechanisms under thermal-shear coupling fields. Therefore, future research could explore the structural evolution patterns of these types of heavy oils under CO<sub>2</sub>-thermal-shear multifield coupling conditions, to further improve the theoretical system of heavy oil development and expand the adaptability boundaries of process technologies.

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## Notes

The authors declare no competing financial interest.

## ACKNOWLEDGMENTS

The authors thank National Key Research and Development Program (2024YFF0506503), the National Natural Science Foundation of China (Grant No. U23b6003), and the Postdoctoral Fellowship Program (Grade B) of China Postdoctoral Science Foundation (GZB20250681) for the support of this work.

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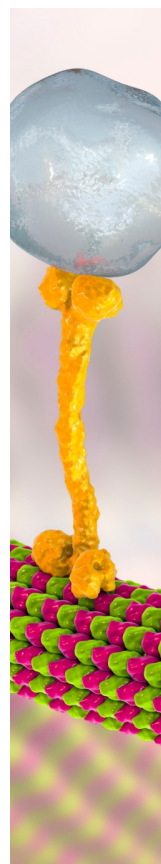
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